

Macroeconomic Model for Bolivia Using System Dynamics

I. Introduction

System dynamics models were introduced in the 1960s and developed by J. Forrester at MIT. They have been applied in different branches of science, technology, and, above all, economics. As their name implicitly indicates, these models make it possible to model complex systems by transforming their elements into an arrangement of flows and stocks, using control variables and relationships that reflect their interactions through simple equations. These equations lead to the construction of systems of differential equations, taking into account that a stock is the integral resulting from the difference between inflows and outflows, which in turn represent the rate of growth of that integral; in a physical analogy, this rate of change is represented by a derivative. This document does not intend to constitute a treatise on system dynamics, but rather to emphasize the use of these tools for the construction of a model of the Bolivian economy that, through calibration and improvement, could become a tool that allows simulation to be used as a policy experimentation laboratory.

II. Background

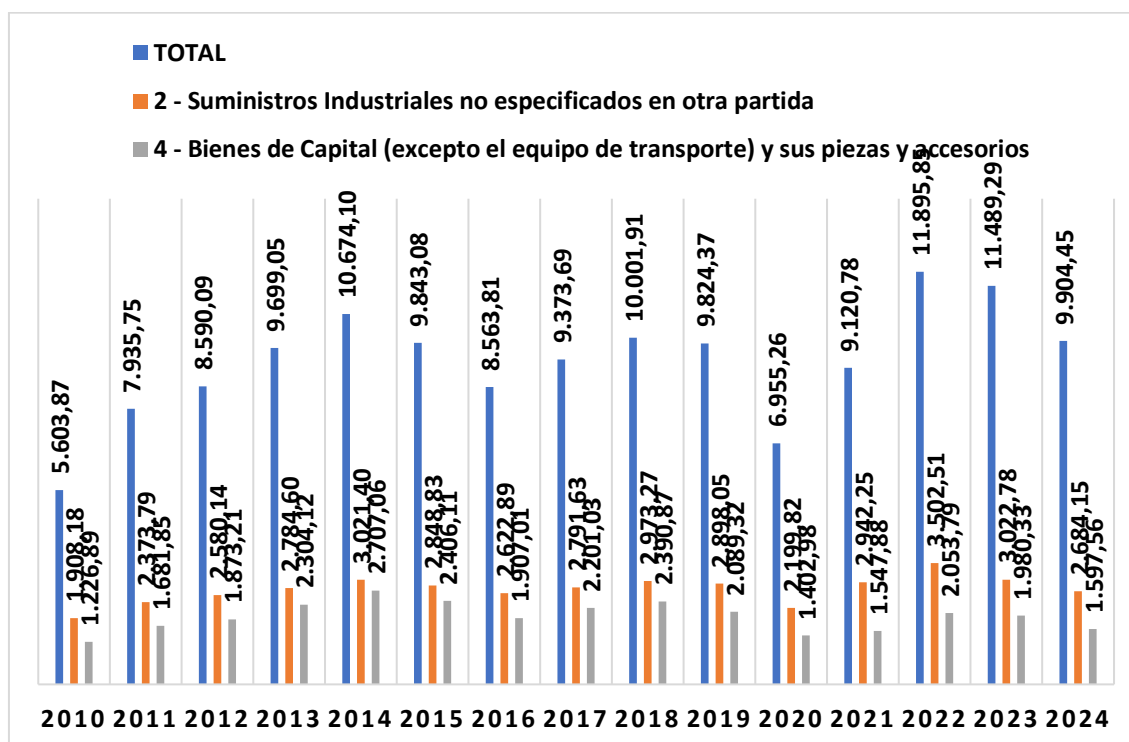
The Bolivian economy is a relatively underdeveloped economy, with limited use of technology and, especially, high technology; deficiencies in technical and technological training; and low productive integration. It bases its foreign currency generation on the export of raw materials whose prices are set in international markets and which, by their nature and low added value, are highly volatile. Shocks caused by fluctuations in these prices deeply affect the Bolivian economy, and this scenario has persisted since the colonial period, through republican life, and in recent years under the current Plurinational State.

From the supply perspective, national production, in addition to having a modest stock of capital and low levels of technical and technological training that limit the capacity of the labor factor, shows a disconnection among productive sectors. Bolivia is highly dependent on imports of machinery and industrial equipment and, mainly, on inputs and processed raw materials for industry, such as steel, polymers for plastic production, cellulose, basic chemical industry products, and pharmaceutical inputs. The Bolivian industry has not been able to take off in the country despite the availability of a large share of the necessary raw materials. Therefore, a fall in the terms of trade, mainly due to raw material prices, together with the decline in our gas reserves, generates pressure toward the devaluation of our currency. This is associated, in turn, with the decline in international reserves, which makes imports more expensive and creates shortages of the vital inputs needed for national production, which in general remains essentially a maquila-type system.

Finally, the problem is exacerbated by dependence on imported fuels, which in recent years have been priced high due to different crises within the framework of policy.

The following graphs show imports of productive equipment and industrial inputs, as well as imports of industrial inputs compared with industrial GDP.

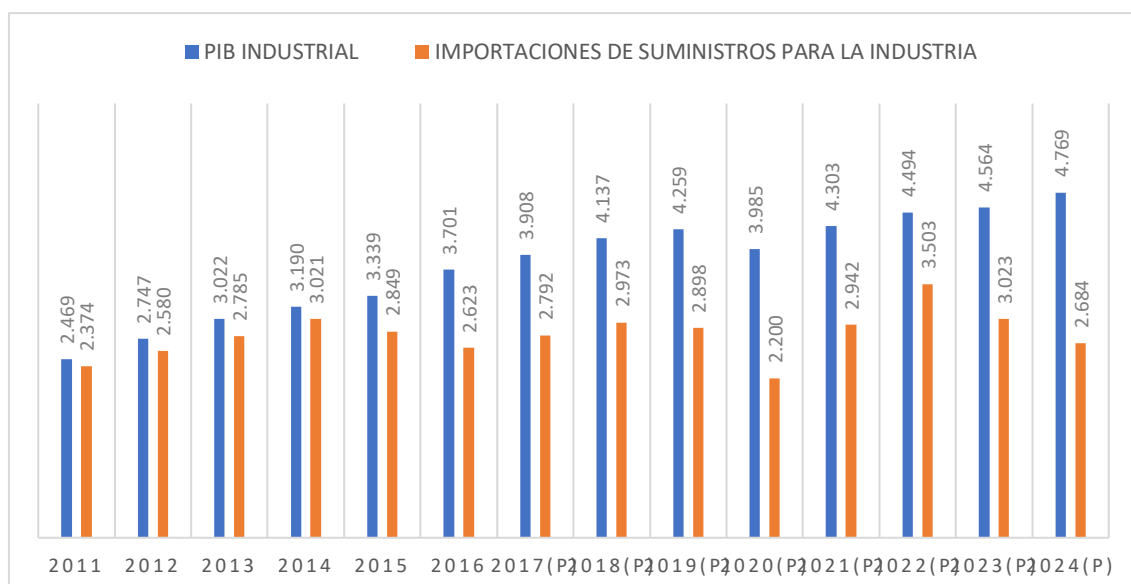
Graph 1. Imports of capital goods and industrial supplies relative to total imports, in millions of USD



Source: Own elaboration based on INE data

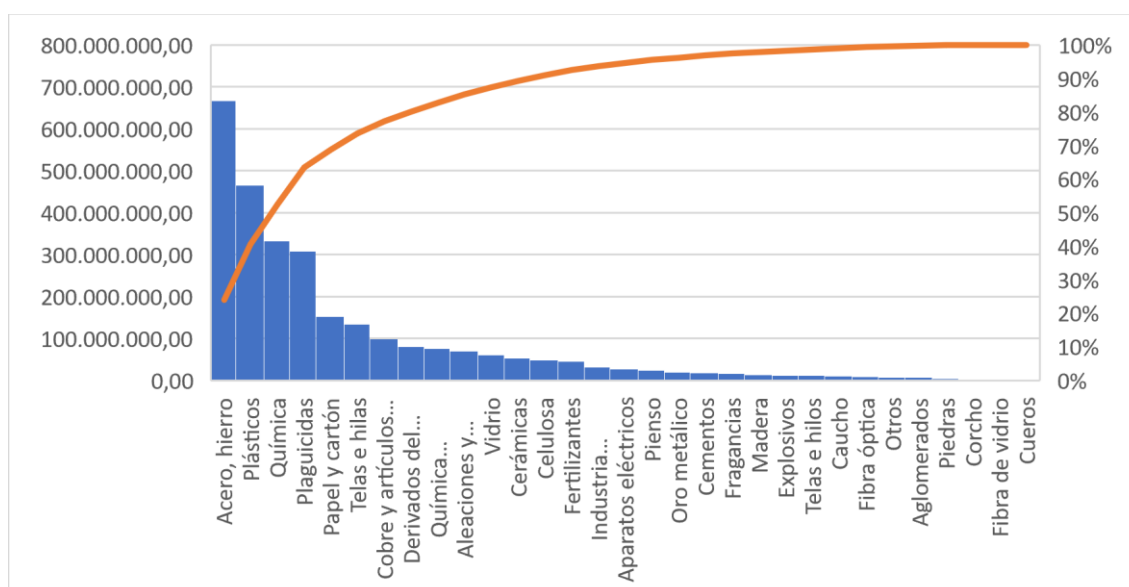
Graph 2.

Graph 2. Imports of industrial supplies compared with industrial GDP, in millions of USD



Source: Own elaboration based on INE data

Graph 3. Imports of the main industrial inputs, in US dollars



Source: Own elaboration based on INE data

With regard to other important aspects, the Bolivian economy is fundamentally informal. As a result, tax pressure is concentrated on a small share represented by the formal economy, which limits its expansion relative to its potential. Above all, informality is behind the generation of vulnerable and low-quality employment, which weakens the economy from the demand side and therefore weakens social welfare, which is the ultimate objective pursued. Although inequality has been easing in recent years, as can be observed in the evolution of indicators such as the Gini index, major differences remain: a middle class that, although it has grown, is still highly vulnerable, and a significant population group living below poverty thresholds. This is the main source of social conflict, and such conflict complicates economic activity through road blockades, strikes, and protests.

It is also necessary to refer to the serious logistical problems faced by the country, not only related to its landlocked condition, but also to limited infrastructure and information management systems. Nevertheless, with an intelligent integration policy, Bolivia could make better use of its geographical position through major projects that are no longer under discussion, such as the bi-oceanic train or the development of river transport, dry ports, and logistics infrastructure. This would also require strengthening the use of information management technologies and capitalizing on the opportunities represented by Brazil's need for greater connection with the Pacific and, therefore, China, as well as Peru's port development, among others.

Finally, it is appropriate to comment on weak international integration, with a chaotic policy - or the absence of one - regarding the development of trade instruments that, under a strategic vision, would seek to use these mechanisms intelligently to position value-added Bolivian products in market niches. This would take advantage of unique and unrepeatable conditions such as royal quinoa, llama fiber and meat, high-altitude wine, and others, as well as the development of products that incorporate technology, such as robotics and software development.

III. Objectives

To develop an integral model of the Bolivian economy using system dynamics, in order to lay the foundations for the implementation of an economic guidelines laboratory for the evaluation of results.

Although there are works prepared by Bolivian economists in the fields of macroeconomics and energy that have developed models using system dynamics, these are still scarce, and the development of economic models is more closely linked to econometrics, a science that can be integrated into dynamic models for the calibration and adjustment of their equations.

IV. Development

The system dynamics model considers blocks and sub-blocks in order to proceed in a structured manner through arrangements of flows (represented by arrows that depart from and arrive at rectangles), stocks or levels (represented by rectangles) that define the state of the system, variables and constants in text, and arrows that relate constants, variables, and flows corresponding to the model equations. In this context, the model is divided into the following components:

Supply and Production Block

This includes population evolution, the capital stock, and the technological factor used to model economic production through a Cobb-Douglas-type model, in accordance with economic theory.

This block is related to the demand side through investment, which in turn is related to savings and the interest rate for the positive flow of the capital stock.

Demand Block

This includes consumption, government expenditure, savings related to investment, taxes, exports, and imports.

Savings-Investment Block

Savings, as a proportion of income, which also integrates foreign loans and remittances multiplied by the savings factor, represents the positive flow that, depending on the interest rate - in turn related to the availability of bank reserves - enables investment and, consequently, the growth of the capital stock and production.

Fiscal Deficit Block

This includes government expenditure as an economic policy decision variable determined by the government (in Bolivia, public expenditure was one of the economic policy decisions of fundamental importance due to the fixed exchange rate), subsidies, taxes, and part of hydrocarbon exports. This stock influences the Bolivian economy mainly through the loss of value of the boliviano, due to the need to finance part of it through currency issuance.

External Debt Block

This includes foreign loans and debt-service payments, which are important because of their influence on the stock of international reserves, a key determinant of the exchange rate in the national economy.

Trade Balance Block

This includes export income, remittances, loans, external debt service as a proportion of accumulated debt, and imports. This is a key stock in the national economy because it determines the value of the national currency and influences costs, availability for production, and inflation.

Money Supply and Inflation Block

The evolution of the money supply considers the normal requirement of the economy, as well as the requirement related to the sale of dollars to the public and the issuance of currency to finance a proportion of the fiscal deficit.

Conditions for Modeling

Time Horizon

The economy will be considered from 2021 onward, over a 50-year time horizon, with a time step of one year, for ease of figure calibration, although it could be modeled semiannually or quarterly for greater precision.

Figures

Figures will be used at constant and proportional prices, with 1990 as the base year, considering increases in the price level relative to 2021 in order to simulate the effect of inflationary pressure on the evolution of the economy.

Other

Values corresponding to foreign currency will be calculated in terms of constant 1990 prices.

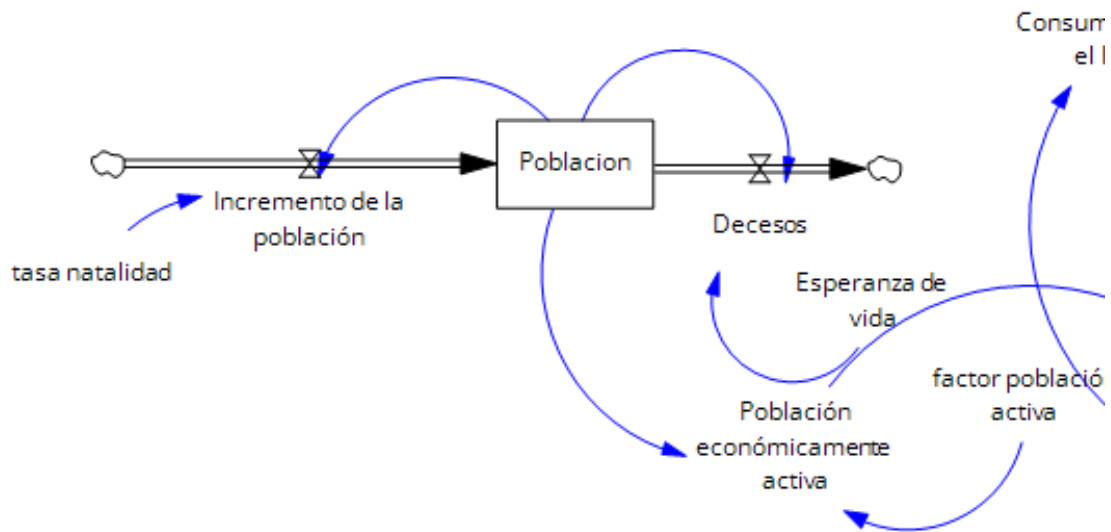
The development of the model in its different components is shown below, in accordance with the above.

Population Evolution

According to the latest population census, the final results of the INE established 11,365,333 inhabitants, with an intercensal rate relative to the last census in 2012 of approximately 1.2% per year.

The flow and stock diagram shows the model as indicated in the figure.

Figure 1. Population Evolution



$$\frac{d(Población)}{dt} = \text{incremento de la población} - \text{Decesos}$$

$$\text{Incremento de la población} = Población * \text{tasa de natalidad}$$

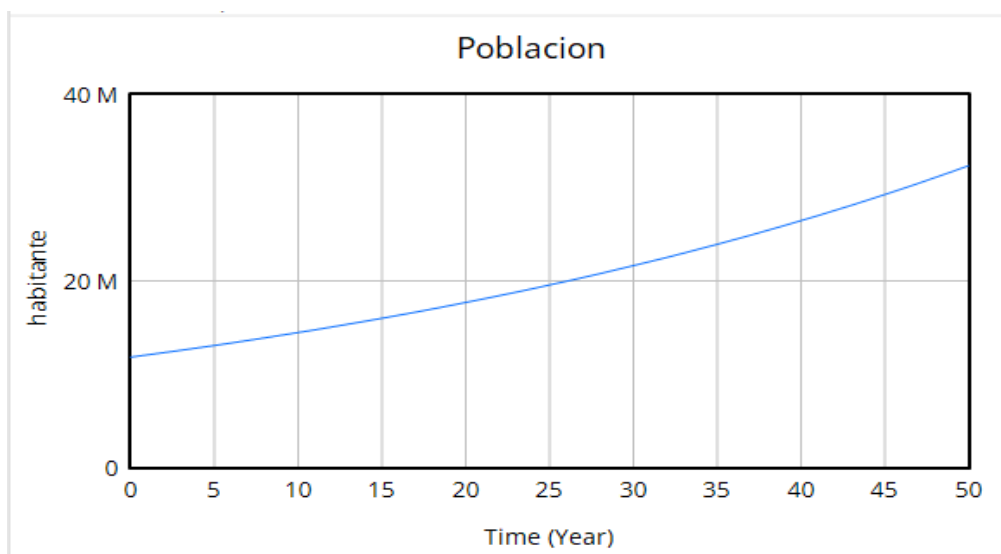
$$\text{Decesos} = \frac{Población}{\text{Esperanza de vida}}$$

$$\text{población económicamente activa} = Población * \text{factor población activa}$$

Birth rate = 3.42%, life expectancy = 74 years, active population factor = 0.7

These data are consistent with the Bolivian economy, resulting in the projection shown below:

Fig. 2. Population Evolution



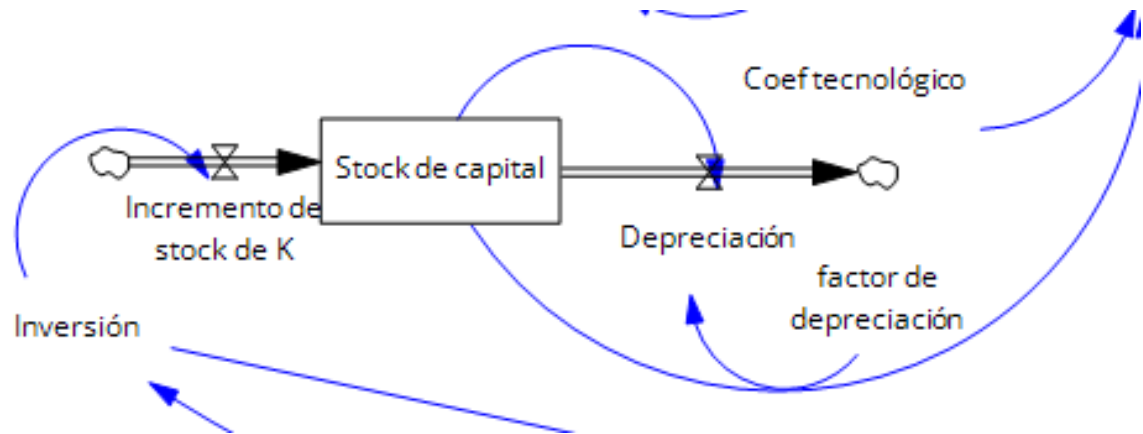
National Production

For this block, a Cobb-Douglas model of the following form will be used:

$$Producción = A * K^{\alpha} * Población^{1-\alpha}$$

In this context, the evolution of the capital stock K is associated with the following model:

Fig. 3. Evolution of the Capital Stock



$$\frac{d(\text{Stock de capital})}{dt} = \text{Incremento de stock de capital} - \text{depreciación}$$

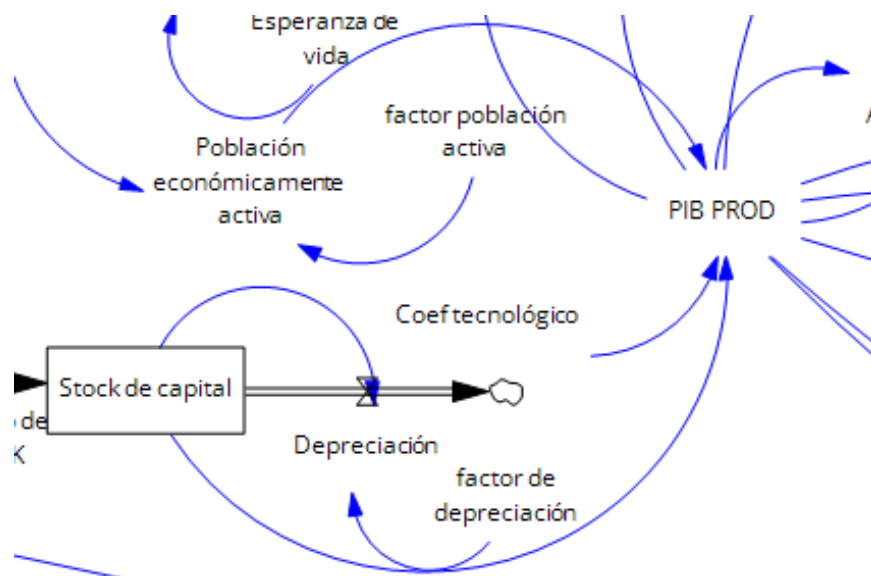
$$\text{Incremento de stock de capital} = \text{inversión}$$

$$\text{Depreciación} = \text{Stock de capital} * \text{factor de depreciación}$$

Where: depreciation factor = 1% (this value corresponds to a calibration carried out for the Cobb-Douglas function)

GDP is therefore modeled as follows (see incidence arrows):

Figure 4. GDP Model



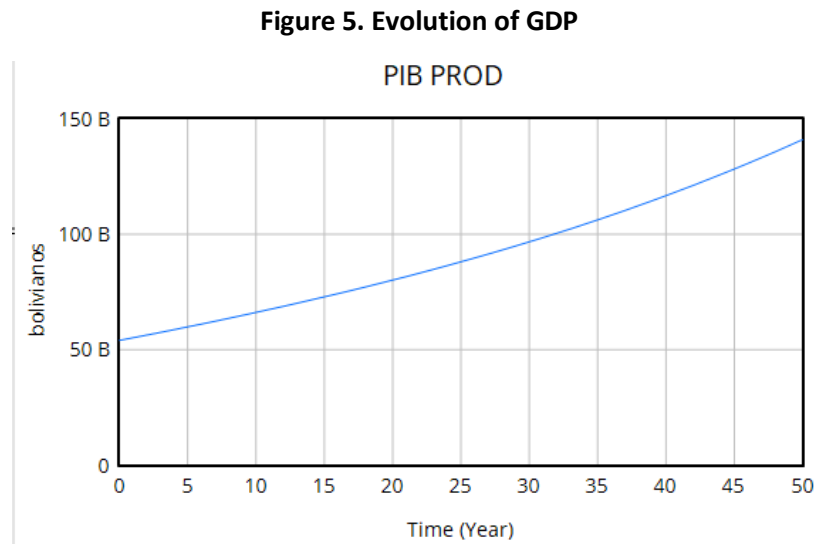
$$PIB PROD = Coef * Stock de capital^{\alpha} * Población activa^{1-\alpha}$$

For which:

Coef: result of a calibration and adjustment of the equation for GDP growth under normal conditions, close to 2%; for this model it is 144.

Alpha: equal to 0.35, which is an estimated value for a developing economy.

The modeling of production growth under normal conditions, using these equations, is shown in the following figure:



Growth showing a rate of 2%.

Component Corresponding to Demand

Consumption

Consumption includes the propensity to consume multiplied by GDP plus remittances received, with an approximate annual amount of 1.4 billion dollars adjusted to 1990 bolivianos, applying a factor of 0.8 - a tax rate considered as 0.24 - plus subsidies equal to 2.5 billion 1990 bolivianos.

Savings

Considered as 0.2 of productive GDP, which increases the flow of bank reserves.

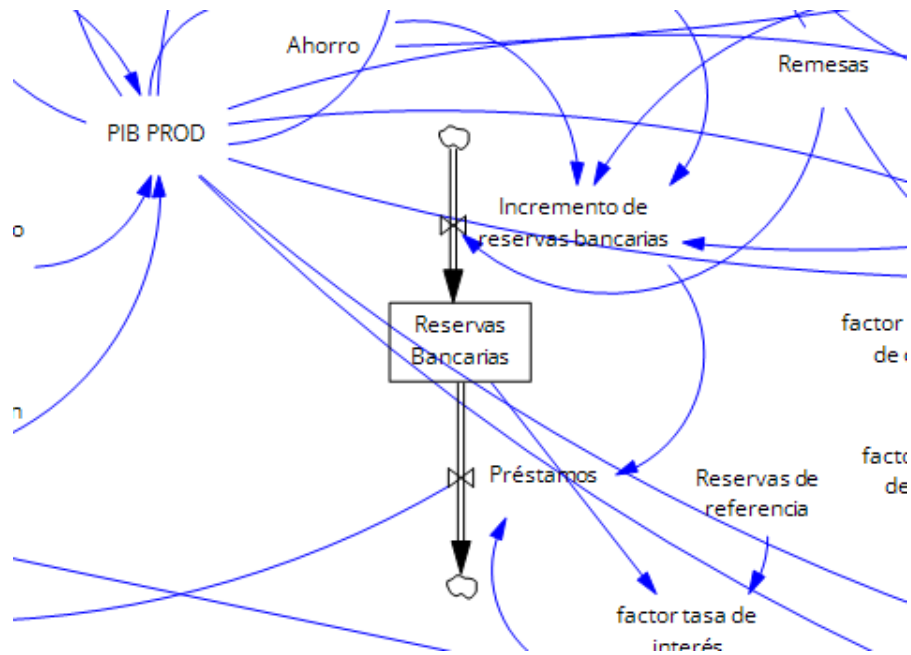
Government Expenditure

At the start of the model, it is considered as 0.3 times GDP, an amount greater than that considered for taxes.

Bank Reserves, Investment, and Interest Rate

With regard to this point, the model is shown in the following figure through the flow and stock diagram:

Fig. 6. Model for the Evolution of Bank Reserves and the Interest Rate



The equations that govern this block are described below:

$$\frac{d(\text{Reservas Bancarias})}{dt} = \text{Incremento de reservas bancarias} - \text{préstamos}$$

$$\begin{aligned} \text{Incremento de las reservas bancarias} \\ &= \text{ahorro} + \text{préstamos del extranjero} \\ &+ \text{remesas por la propensión al ahorro} \end{aligned}$$

$$\text{Préstamos} = \text{factor tasa e interés} * \text{Incremento de reservas bancarias}$$

For simplicity, loans are directed toward investment.

The interest-rate factor arises from a nonlinear relationship in a graph associated with the quotient between the stock of bank reserves and the reference reserves (considering an amount approximately equal to one third of the GDP of the first simulation year). This factor increases with the indicated ratio if it is greater than zero and decreases with a ratio lower than zero, as a proxy for the interest rate, considering that:

$$\text{Inversión} = f(\text{tasa de interés})$$

$$f' \left(\text{Reservas} \frac{\text{Bancarias}}{(\text{reservas de referencia})} \right) \geq 0 \quad f'' < 0$$

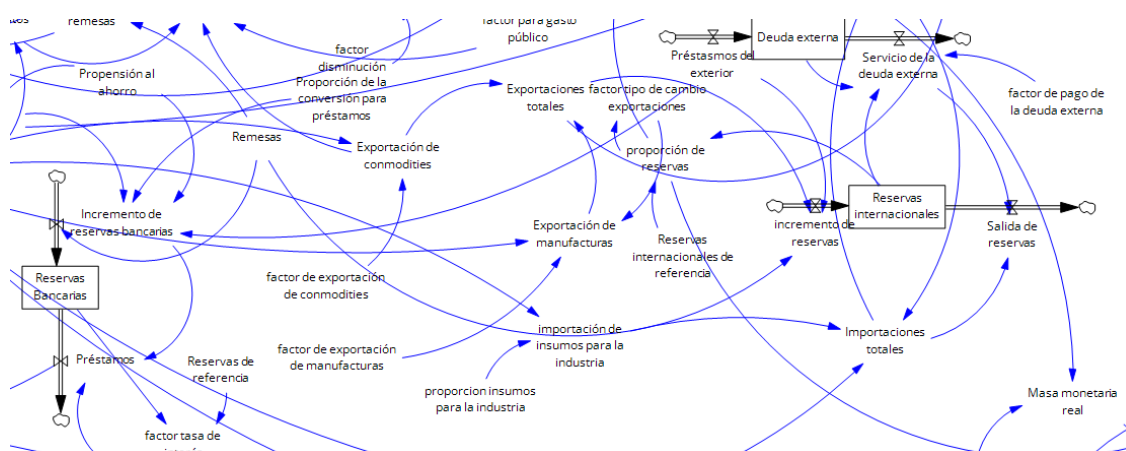
The above relationship indicates that if bank reserves increase relative to a reference rate, the interest rate will fall and stimulate investment, which must be related to the availability of funds (savings, foreign loans, part of remittances, and the reserves themselves). However, this rate may be influenced by central bank policies through reserve requirements and other instruments.

Foreign investment will be added later for scenario analysis.

External Debt and International Reserves

For this sub-block, the following model is considered:

Fig. 7. Evolution of Bank Reserves and the Interest Rate



Imports

Imports are composed of capital goods, as part of total investment for production, estimated at 50%; inputs for industry; and consumer goods, the latter mainly affected by the exchange rate. They are separated because this would allow for better work with scenarios, such as an industrial policy for import substitution of industrial inputs. Imports have the effect of reducing international reserves, whose level in turn affects the real exchange rate and therefore the relationships that make up the trade balance.

The exchange-rate impact factor is related to the decrease and increase of international reserves, with this relationship being inverse.

Imports = imports for consumption (as 40% of consumption) + imports of industrial inputs (as a proportion of GDP calibrated at 7%, according to import data for this category, which represents close to 3 billion current dollars).

Exports

On the other hand, exports are composed of manufactures, which are mainly affected by the exchange rate in an inverse manner, and commodities, which are mainly affected by the level of international prices. Nevertheless, for the sake of model simplicity, it will be assumed that half of these correspond to the total, but as a proportion of GDP. Exports positively affect net international reserves, as do remittances and foreign loans.

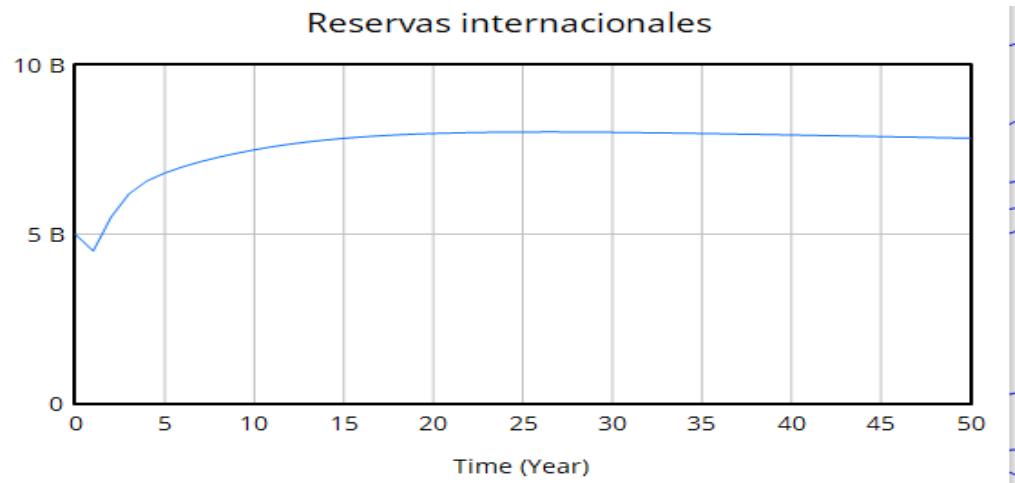
International Reserves

These are increased by foreign loans, which will initially be assumed at 300 million dollars adjusted to 1990 prices; remittances from abroad, close to 1.4 billion dollars per year; and exports, as indicated above. They are reduced by imports, as explained above, as well as by debt service calculated as a fraction (10%) of the accumulated value of external debt that

must be honored periodically, in this case annually. From an orthodox perspective, this factor should be related to the country's risk rating, an aspect not addressed in detail in this model in order to keep it reasonably simple.

For the model with the initial conditions, the following evolution is shown, which considers inertial equilibrium.

Fig. 7. Evolution of International Reserves

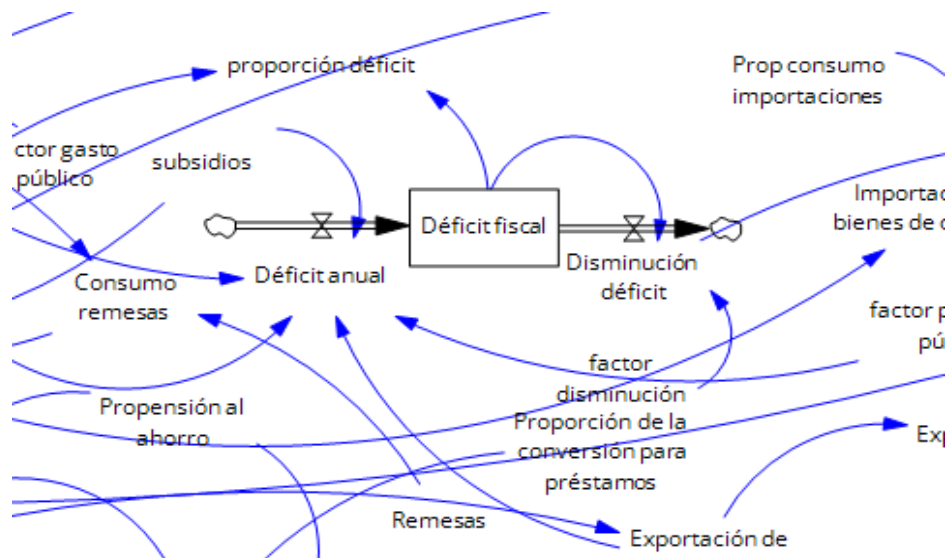


Subsequently, in the scenario analysis, a disturbance related to the fall in commodity exports will be shown, mainly gas, which has led to a decline in reserves, a larger fiscal deficit putting pressure on the real devaluation of the currency, and an increase in price levels due to the effects of money supply and the exchange rate.

Fiscal Deficit

The block of the model related to the fiscal deficit is shown in the following figure:

Fig. 7. Model for the Evolution of Bank Reserves and the Interest Rate



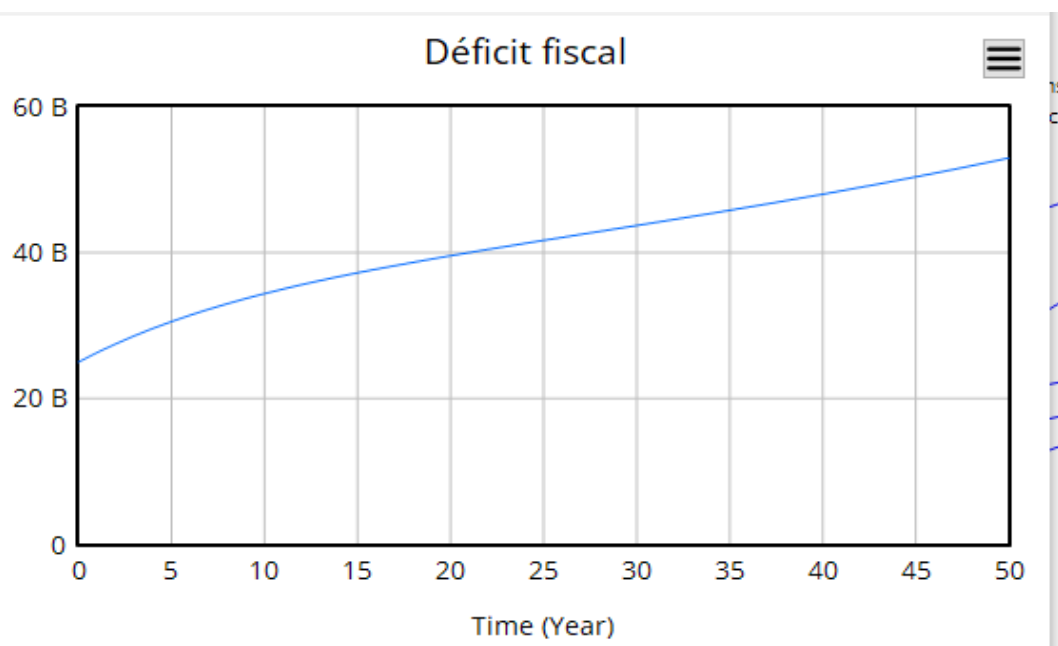
The equations and relationships are described below:

$$\frac{d(\text{Déficit fiscal})}{dt} = \text{Déficit anual} - \text{disminución déficit}$$

The model initially starts with a deficit of 30% of GDP, and this accumulates through the annual deficit generated by the difference between public expenditure related to GDP and subsidies - considered as an independent variable for modeling purposes - minus taxes and minus gas export revenues. The reduction of the deficit is equal to a proportion of the accumulated deficit, as a government policy, which at the start of the modeling has been considered as a factor of 10%.

The initial modeling shows the following evolution.

Fig. 8. Evolution of Bank Reserves and the Interest Rate



If the proportion relative to GDP is considered, the following graph is shown.

The fiscal deficit affects the economy through the need to finance it by issuing currency, which increases the money supply and puts pressure on inflation, or through operations that increase public debt and the commitment to pay interest. The accumulation of the fiscal deficit is directly proportional to public debt.

Money Supply and Inflation

With regard to this block, the evolution of the money supply is modeled as the relationship between the portion that accompanies GDP (as 30% of it) and monetary issuance, which also includes fiscal deficit financing as the monetary flow defined by the government for its reduction, according to the section explained above.

This sub-block has not been developed in greater detail in order to keep it simple, since it could be considered as a stock and include operations to withdraw and make available the money supply through the purchase and sale of securities issued mainly by the Central Bank.

Finally, inflation is modeled with an index that has the following equation, as a weighted average between the effect of the money supply, the exchange rate, and GDP on the demand side (consumption, subtracting taxes + savings/investment + government expenditure + exports - imports):

Indice de precios

$$= \left(\alpha * \left(\frac{\text{masa monetaria total}}{\text{masa monetaria relacionada con el PIB}} - 1 \right) + \beta * \left(\frac{\text{Reservas de referencia}}{\text{reservas internacionales} - 1} \right) + \gamma * \left(\frac{\text{PIB Demanda}}{(\text{PIB Producción}) - 1} \right) \right) - 1$$

This is a factor that deflates GDP calculated at constant prices.

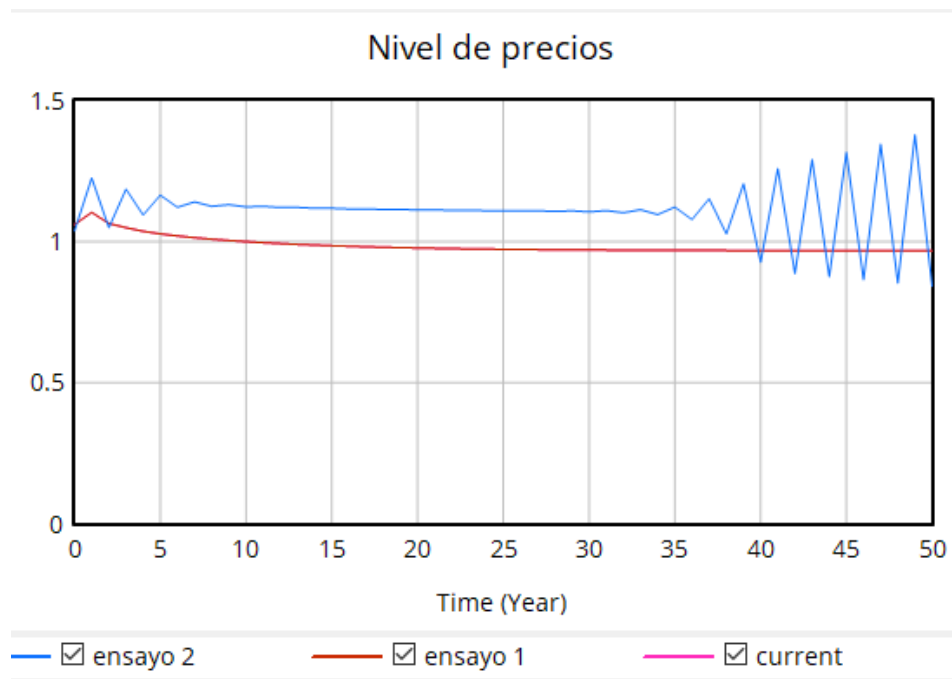
The evolution of this index under normal conditions is asymptotic, with a tendency toward 1.

Model Disturbances and Scenario Analysis

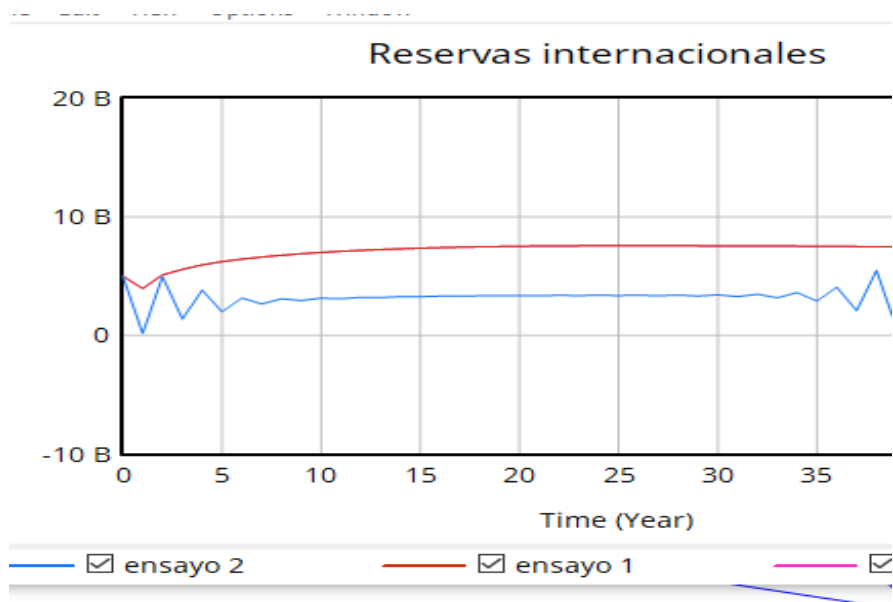
As a first scenario, a fall of half of commodity exports is analyzed. In this case, they are mainly affected by the decline in gas, considering that a fraction of them (those corresponding to gas) forms part of public revenue, thereby affecting the fiscal deficit and inflationary pressure due to the need to finance it through currency issuance and through the exchange rate as the positive flow of international reserves decreases.

In this context, the model results are as follows:

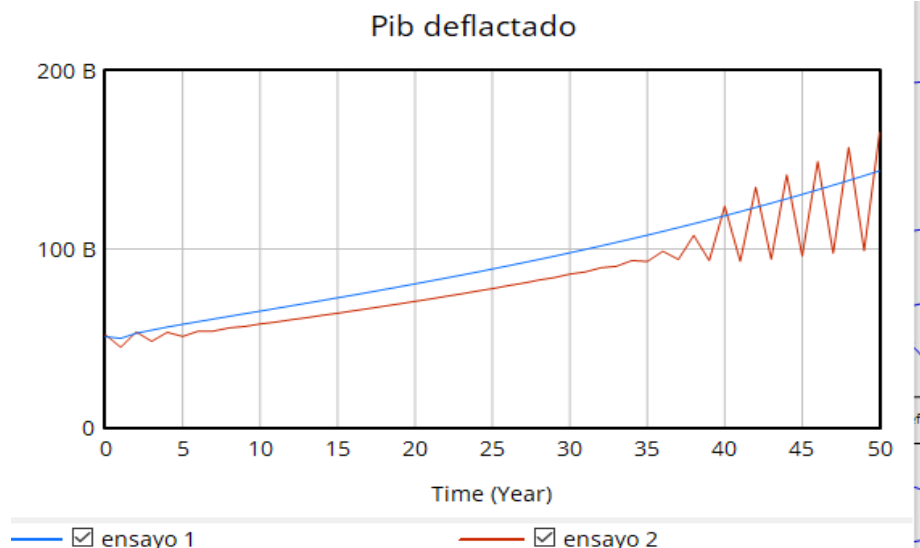
Scenario 1



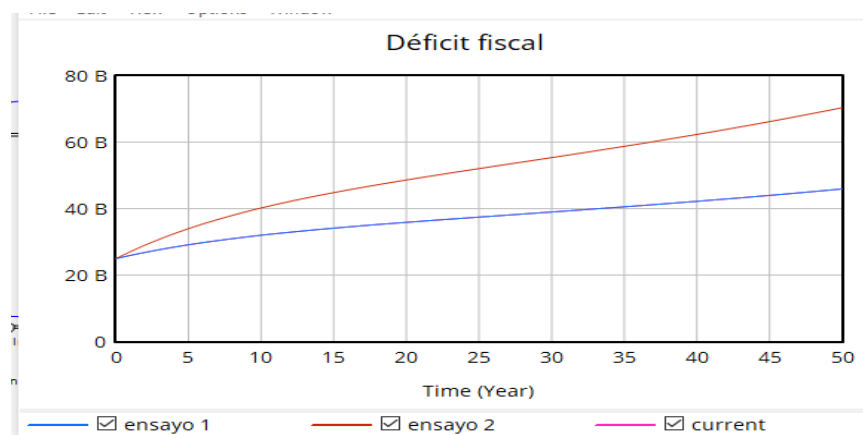
An increase in the price level is evident, which in equilibrium should be one, taking into account that the model uses constant prices (blue line).



A proportionally decline in international reserves, blue line.

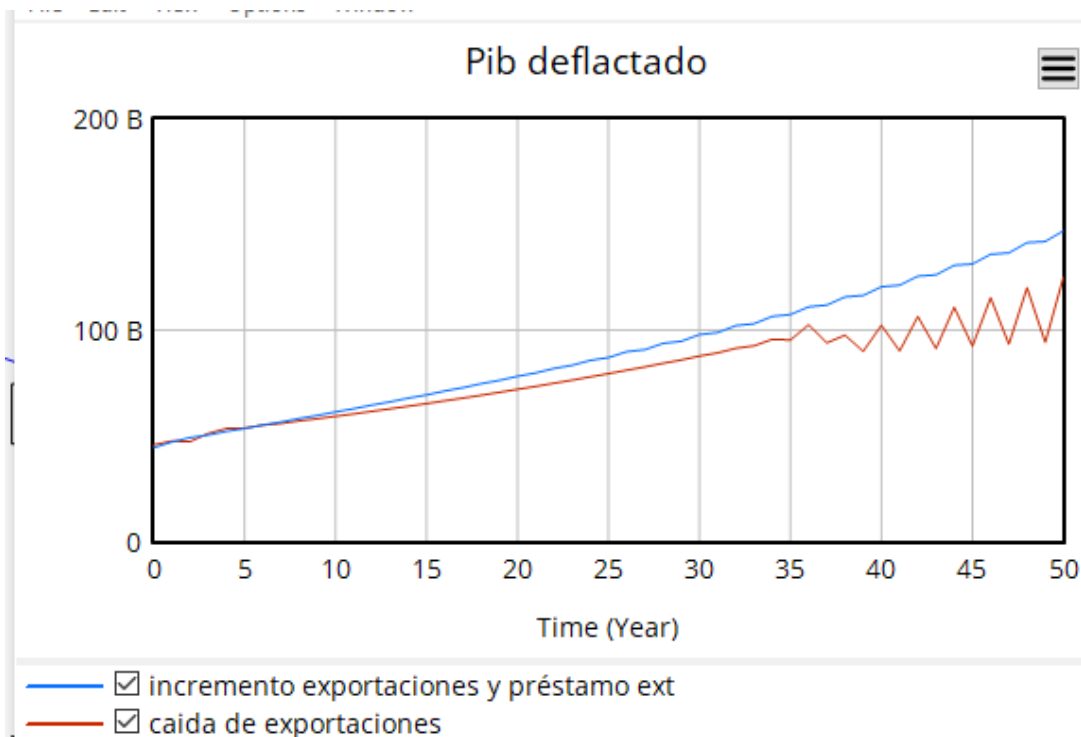


A real fall in GDP, red line (trial 2), due to the increase in the price level and the fall in exports.



Increase in the fiscal deficit.

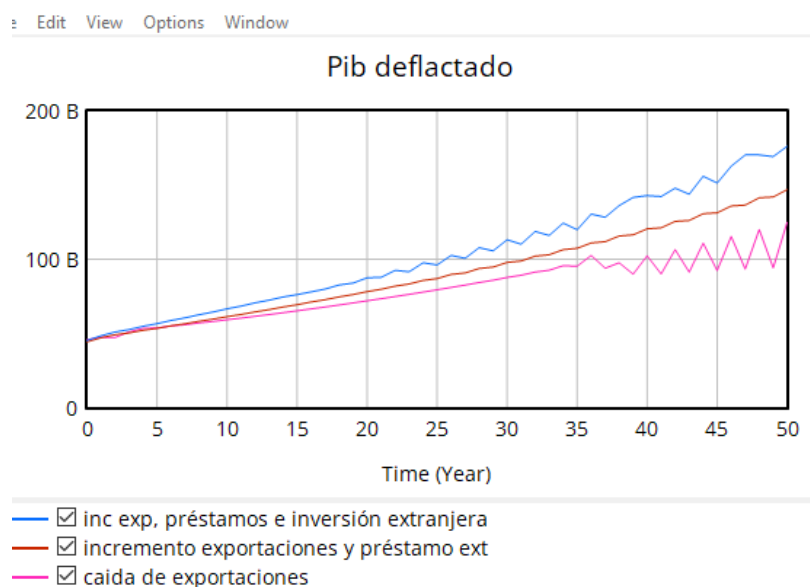
In a second scenario, the increase in exports and the increase in domestic financing through foreign loans to increase investment capacity are shown in the following graph.



A sustained increase in GDP can clearly be observed, although the model shows limitations, especially in relation to the Cobb-Douglas function, where the Alpha coefficient could even be increased for a greater incidence of capital and, therefore, investment in this category.

Scenario 3

In this scenario, foreign investment is included as 10% of GDP, producing the following result:



Highlighting higher projected growth.

Conclusions

A model has been built using system dynamics tools, integrating theoretical macroeconomic models and concepts applied to the Bolivian context, considering the different components of the economy and their interrelationships that make it possible to project scenarios. Nevertheless, it is a model that would require adjustments and the inclusion of mathematical and causal relationships to expand it and improve its precision so that it can become a simulation instrument, such as incorporating the evolution of dependence on inputs and processed raw materials through their substitution by national production, the evolution of real wages, and other aspects.

However, it is clear that the deterioration of the Bolivian economy is related to a decline in exports that affects the exchange rate, and through it inflation, by increasing the cost of imported products. It is also related to the increase in the fiscal deficit in the context of very high public expenditure, due to the fall in fiscal revenues from gas exports, which puts pressure on currency issuance, increases the money supply, and raises price levels.

The lines identified for improving the economy are:

- Increase international reserves in order to stabilize the exchange rate, prioritizing their level in the short term.
- Increase manufacturing exports by improving quality, adding greater value, incorporating technology, and developing new industries such as software and tourism, as well as mining and hydrocarbons by increasing exploration while considering their industrialization within the country as far as possible. This must also be accompanied by large-scale logistics development and an appropriate strategy in the area of international relations for the opening and consolidation of markets.
- Open the country to foreign investment, mainly for the development of industries that the country does not have, such as inputs and processed raw materials for industry, especially chemicals, industrial equipment, and polymers. This would strengthen the domestic industrial fabric and requires the design of an aggressive tax policy that generates significant advantages.
- Improve technical and educational levels by strengthening and increasing human capital.
- Improve the country's logistics conditions, both in infrastructure and information technologies, to enable better international integration and take advantage of Bolivia's geographical position at the center of South America.
- Improve the country's capacity for clean energy generation by alternative sources in order to be least dependent of fuel import.

References

Bibliografía

- Jon Sterman, *Business Dynamics: Systems Thinking and Modeling for a Complex World* (2000),
- Donella Meadows, *Limits to growth*, 1972
- *Industrial Dynamics*, Jay Forrester, 1961.
- *Macroeconomía*, Samuelson Davidson, 1990.

- Macroeconomy, Gregoy Mankiw, 2006.
- Instituto Nacional de Estadística.
- La función de producción Cobb Douglas y su aplicación en la economía boliviana, Dante Ayaviri Nina, Pedro Feraudi Gonzáles.